6TH GRADE MATH · UNIT 4: EXPRESSIONS AND EQUATIONS

Arcade Party Planner

Write expressions and solve equations to plan a birthday party at the arcade.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your cousin wants a birthday party at Pixel Palace Arcade. Each guest needs a game card, the party room costs a flat fee, and prizes are bought with tickets.

You will write expressions and solve equations to compare party packages and plan a party that fits the budget.



Party Room

$40 flat fee

up to 3 hours

Game Card

$12 per guest

1 credit = 1 game

Learning Goals

- I can write and evaluate expressions with variables and exponents.
- I can use properties to write equivalent expressions.
- I can write and solve one-step equations in context.

Project Timeline

When	What you will do
Day 1	Package Expressions
Day 2	Equivalent Expressions
Day 3	Equations & Design
Final	Party Package Proposal + reflection

Part 1: Package Expressions

Day 1 · Let g be the number of guests. Use the Arcade Price Board.

Arcade Price Board

Game Card	Pizza	Drinks
$12 per guest	$15 per pizza	$2 per guest
60 credits	serves 4	

Party Room	Prize Tickets	Cake
$40 flat fee	Ticket bonus: 50	$25 per cake
	per guest	

1. Write an expression for the total cost with the room, a game card and a drink for each guest, and one pizza for every 4 guests.

2. Evaluate your expression for 8 guests and for 12 guests.

3. Pixel Palace also offers "The Mega": $20 per guest, everything included, no room fee. Write its expression and evaluate it for 8 and 12 guests.

4. Which package is cheaper for 8 guests? For 20 guests? What do you notice?

Part 2: Equivalent Expressions

Day 2 · Use properties of operations.

5. Use the distributive property to write 6(2g + 5) without parentheses. What could it describe?

6. Factor 18g + 24 using the greatest common factor.

7. Combine like terms: 12g + 40 + 2g + 25 + 3g.

8. A super prize costs 3^4 tickets and a mega prize costs 4^3 tickets. Which costs more and by how much?

Part 3: Equations & Design

Day 3 · Write and solve an equation for each question.

9. The Mega package cost $260. How many guests came?

10. After the room fee was paid, the family had $186 left for game cards and drinks at $14 per guest. How many guests can come?

11. Design your own package with a price per guest and a flat fee. Write its expression and an equation for a $300 budget. Solve it.

Final Product: Party Package Proposal

Your final product must include:

- Expressions for both packages and your own
- A table comparing costs for 8, 12, and 20 guests
- Two equivalent expressions with the property used
- Equations solved for guests and budgets
- Party proposal with a recommendation

Reflection

R1. Why do the regular and Mega packages switch which is cheaper?

R2. How do you know two expressions are equivalent?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Expressions and equations model every cost correctly and are used to compare packages and plan within the budget. Goes beyond the minimum with a creative, realistic choice.	Expressions and equations model every cost correctly and are used to compare packages and plan within the budget.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes expressions, a comparison table, solved equations, and a clear recommendation. Polished, organized, and visually engaging.	Proposal includes expressions, a comparison table, solved equations, and a clear recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.